Javier Aldana

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EDUCATION

Purdue University — M.S. Graduation Date: December 2027
B.S. & M.S. in Electrical Engineering — West Lafayette, IN

- GPA: 3.7/4.0
- Relevant Coursework: Power Electronics, Robotic Systems, Microprocessors, Feedback Systems & Controls, Advanced C, Data Science, Digital Signal Processing, Electromagnetics

PROFESSIONAL EXPERIENCE

ASML – Wilton, CT — May 2025 – Aug 2025

Production Engineering Electrical Support & Test Intern

- Designed PCB for automated electrical box to qualify resistance checks on ASML lithography machine. The board included pin selection, internal calibration, and measurement accuracy within ±.1Ω. The integrated box now reduces production cycle time during fabrication qualification, **saving ~$45,000 annually**.
- Resolved cleanroom escalations by rewiring, probing, detecting EMI, diagnostics, consolidating data, ECab relocation, and running firmware updates on cleanroom equipment.

Purdue Space Program Active Controls – West Lafayette, IN

Lead Avionics Engineer — Jan 2025-Present

- Designed DAQ/Valve Control board for lander rocket TOAD, with STM32H7 MCU, analog filtering, ADC, CAN, SPI, RS485 communication, and high-power switching.
- Owned the development of house-made 40kV Spark box to test team-designed torch igniter.
- Directing design of custom four voltage level Buck Converter PCB on Altium.
- Built and integrated DAQ PCB to collect real-time pressure data on our liquid rocket engine TADPOLE, completing the Throttle Milestone for the Collegiate Propulsive Lander Challenge, winning **$15,000**.
- Assembled and laid out DO & AI lines for Test Stand DAQ system at Zucrow Labs.

Purdue Mechanisms and Robotic Systems Lab - West Lafayette, IN — Sep 2025 – Dec 2025

Undergraduate Research Assistant

- Developed vision-based tactile sensor for robot, integrating real-time RGB data to give haptic feedback to robotic arm using Point Cloud algorithm.
- Manufactured several sensor components including, gels, housings and lenses.
- Ensured consistent experiment functionality by diagnosing and repairing various sensors, via precision soldering, wire harness assembly.

Polytechnic University of Madrid - Madrid, Spain — May 2024 – July 2024

Undergraduate Research Assistant

- Set up and wired UNITROL1000 voltage regulator to control and run diagnostics of a 4900 kVA synchronous generator for the research team experiments.
- Debugged electrical wiring for the generator and its relay, rewired and updated the schematics.

SKILLS

- **Programming Languages:** C, SystemVerilog, Python, Java, MATLAB, RISC-V Assembly, Simulink, Embedded C, ROS
- **Electrical Design:** SPICE, Altium, KiCAD, Arduino, FrontPanelExpress
- **Electronics & Hardware Skills:** Circuit Analysis, Soldering, Cable Harnessing and Assembly, Oscilloscope, Signal Generators, NI TestStand, 3D printing